CHE374: Economic Analysis and Decision Making

Lecture Notes · Weeks 1 to 5

Rhyl · Fall 2026

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1 Lecture One: Time Value of Money

Concept of time value of money: borrowing money has a cost for the borrower, and lending money should create value for the lender. You get back the original money plus interest, and the **interest rate** is

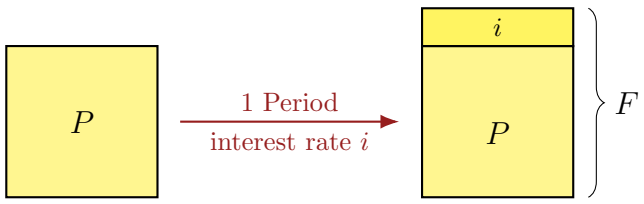
$$i = \frac{\text{interest}}{\text{principal}}.$$

Notation:

- P: present value
- F: future amount
- i: interest rate

So over one period:

$$F = P + Pi, \qquad \Longrightarrow \qquad F = P(1+i).$$



One period growth: the principal P earns interest Pi, giving $F = P(1+i)$.

Interest rates (in %)

The rate depends on the type of investment:

1. **T-Bill** (short term loan to the Bank of Canada): typically earns the lowest return but is considered risk free.
2. **Bonds** (lending money to a provincial or Canadian government): pay a slightly higher interest rate.
3. **Personal loans or loans to small businesses:** riskier, but a significantly higher interest rate.

Factors that affect the rate:

1. Inflation
2. Credit (default) risk
3. Liquidity risk
4. Maturity

Simple vs. compound interest

Notation again:

- P: present (principal)
- F_N: future amount in year N
- i: interest rate
- I: interest amount

Simple interest applies only to the original principal:

$$F_N = P(1 + Ni).$$

Compound interest applies to the principal and all interest accrued so far (e.g. mortgages, credit card debt):

$$F_N = P(1 + i)^N.$$

For the same rate r per year, the more frequently you compound (the bigger m) the more you earn.

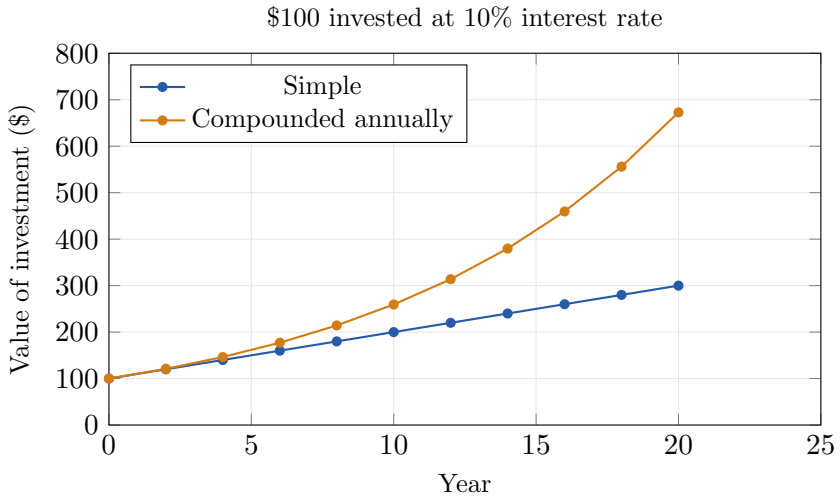
Effective interest rate

The **effective interest rate** is the equivalent interest rate of an account that is compounded just once over a year:

$$i_e = \left(1 + \frac{r}{m}\right)^m - 1.$$

Here we compound multiple times per year:

- r: nominal interest rate (for one year),
- allowed to compound m times (subperiods) per year,
- $F = P\left(1 + \frac{r}{m}\right)^m$,
- $\frac{r}{m} = i_s =$ subperiod rate.



Simple interest grows linearly; compound interest grows exponentially.

Continuous compounding

$$i_e = \lim_{m \to \infty} \left(1 + \frac{r}{m}\right)^m - 1 = e^r - 1.$$

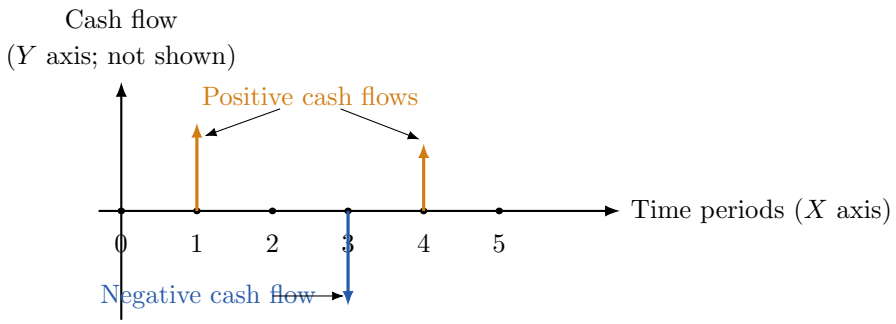
2 Week Two: Cash Flows and Equivalence

Categories of cash flow

1. **At the start:**
   - **First cost** (capital cost).
2. **During a project (investment):**
   - **Revenue:** we will have some revenues or savings (car example).
   - **O&M** (operation and maintenance): e.g. car insurance or fuel.
   - **Overhaul**, a major expenditure: e.g. an engine requires an overhaul about half way through its life.
3. **At the end:**
   - **Salvage value:** net receipt at the end, e.g. from the sale of a piece of equipment (sell the car).
   - **Scrap value:** value in the materials the item is made of.
   - **Disposal cost:** the cost to dispose of a waste when it is no longer usable.

Cash flow diagrams

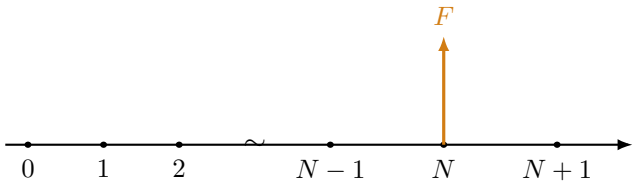
Time 0 is the beginning of period one. A cash flow occurring during a period is accrued at the *end* of that period.



Up arrows are receipts (positive); the down arrow is a disbursement (negative).

Standard cash flow patterns

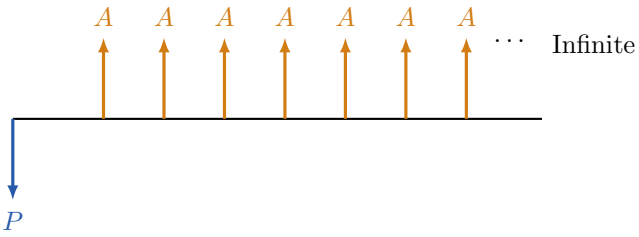
1. Single payment.



A single future amount F received at time N.

2. Perpetuity: an annuity that pays A forever.

$$P = \frac{A}{i}.$$

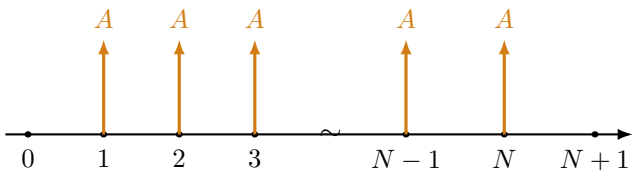


Perpetuity: a level payment A that never ends, $P = A/i$.

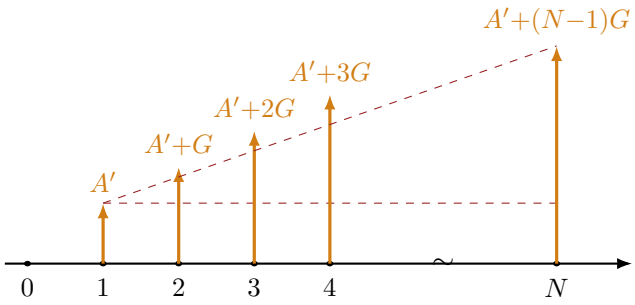
3. Annuity: a perpetuity but only for N periods.

4. Arithmetic gradient. We assume the first cash flow happens at period two, so i starts at period one; we treat the constant first part as an annuity A' and add the growing part G.

5. Geometric gradient: each cash flow grows by a constant factor $(1 + g)$.



Annuity: a level payment A for N periods.



Arithmetic gradient: a base annuity A' plus a linear growth G each period.

Three kinds of equivalence

1. Mathematical equivalence: two cash flows P_t at time t and F_{t+N} (or F_{t+N+M}) are equivalent with respect to the interest rate:

$$P_t = \frac{F_{t+N}}{(1+i)^N}.$$

2. Market equivalence: the idea that one can exchange cash flows between present and future amounts.

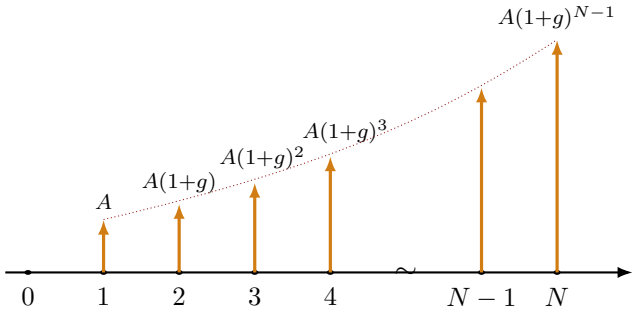
Example. A Treasury Bill (T-bill) is a short term government debt obligation backed by the government, with a maturity of one year or less. T-bills are usually sold in denominations of \$1,000. If a \$1000 T-bill due to mature in 6 months is currently selling for \$990.10, that means some investors holding the T-bill are willing to sell at this price while others are willing to buy at this price. The "market" has agreed that the appropriate interest rate is 1% over 6 months:

$$990.15\,(1+0.01) \approx 1000.$$

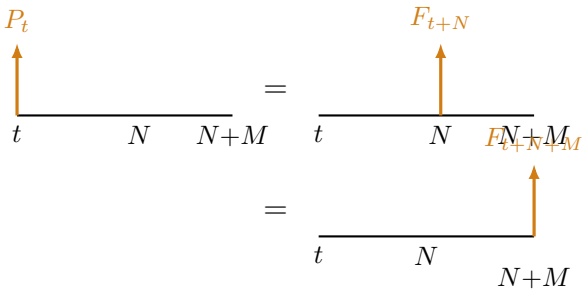
3. Decisional equivalence: indifference of the decision maker among the available choices.

Factor notation

$(X/Y, i\%, N)$, where X and Y are chosen from the cash flow symbols P, F, A, G. If you have Y multiplied by a factor, you get the equivalent value of X, e.g. $P = F(P/F, i, N)$.



Geometric gradient: each payment is $(1+g)$ times the one before.



The same value, seen at time t, at N, and at $N+M$.

$$
\begin{aligned}
(A/P, i, N) &= \frac{1}{(P/A, i, N)} \\
(F/P, i, N) &= (1+i)^N \\
(P/F, i, N) &= (1+i)^{-N} \\
P = A\,(P/A, i, N) &= A\left(\frac{1}{i} - \frac{1}{i(1+i)^N}\right) \\
(P/G, i, N) &= \frac{1}{i^2}\left(1 - \frac{1+iN}{(1+i)^N}\right) = \frac{(1+i)^N - 1 - iN}{i^2(1+i)^N} \\
(P/\text{Geom}, i, g, N) &= \frac{1 - \left(\frac{1+g}{1+i}\right)^N}{i-g}
\end{aligned}
$$

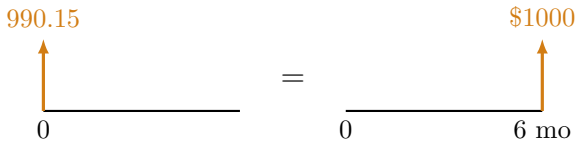
▶ Small corrections to the written factors: continuous compounding gives $e^r - 1$ (not $s^r - 1$); and the (P/G) factor carries $1/i^2$ out front, not i/i^2. Everything else matches the standard factor table.

Or, equivalently, for the geometric gradient:

$$(P/\text{Geom}, i, g, N) = \frac{(P/A, i^0, N)}{1+g}, \qquad \text{where } i^0 = \frac{1+i}{1+g} - 1.$$

Putting the factors to work:

- Annuity: $P = A\,(P/A, i, N)$.



$990.15 today is equivalent to $1000 in six months at 1%.

- Arithmetic: $P = A\,(P/A, i, N) + G\,(P/G, i, N)$. Do the bottom bracket first: the first term is the initial annuity value A that is constant starting at $t = 1$; the second term is the growing value G.
- Geometric: $P = \dfrac{A}{1+g}\,(P/A, i^0, N)$.

3 Week Three: Mortgages and Bonds

Mortgage loan terminology

Buy a house for \$500k = total value.

- **Principal:** you can borrow up to a certain fraction (e.g. \$400k).
- **Down payment:** the rest of it.
- **LTV:** loan to value ratio $= \dfrac{P}{\text{total}}$.
- **Mortgage rate.**
- **Amortization period:** the agreed time horizon for repayment.
- **Term:** the agreed interest rate period (the interest rate can change between terms).

Mortgage Example

These examples did not list the formulas, so they are worked out here, and a general form is given below each part.

\$500,000 home, with a 20% down payment; mortgage rate is 5%, 5 year term, 25 year amortization period.

(a) What is your monthly payment during this term?

The principal is $P = 500{,}000 \times 80\% = \$400{,}000$. Since payments are monthly, use a monthly rate and a monthly count:

$$i = \frac{5\%}{12}, \qquad N = 25 \times 12 = 300.$$

$$A = 400{,}000\,(A/P,\ \tfrac{5\%}{12},\ 300) = 400{,}000 \times \frac{i(1+i)^{300}}{(1+i)^{300}-1} \approx \boxed{\$2{,}338 \text{ per month.}}$$

General formula

$$A = P\,(A/P,\ i,\ N), \qquad i = \frac{r_{\text{mort}}}{m}, \qquad N = Y_{\text{am}} \times m.$$

where A = payment per period; P = principal = (home value)$\times$(1$-$down payment fraction); i = periodic rate; r_{mort} = mortgage rate (annual, nominal); m = payments per year ($m = 12$ for monthly); Y_{am} = amortization period in years; N = total number of payments.

(b) After the end of the term, the new mortgage rate is 6%. What is your new monthly payment?

Two steps. First find the amount still owed after 5 years (that is, after $5 \times 12 = 60$ payments). Take the original principal forward to year 5 and subtract the future value of the payments already made:

$$\text{Amount owed} = 400{,}000\,(F/P,\ \tfrac{5\%}{12},\ 60) - 2338\,(F/A,\ \tfrac{5\%}{12},\ 60) \approx \$354{,}320.$$

This is the principal still outstanding at year 5. Now spread that balance over the remaining $20 \times 12 = 240$ months at the new rate of 6%/12:

$$A_{\text{new}} = 354{,}320\,(A/P,\ \tfrac{6\%}{12},\ 240) \approx \boxed{\$2{,}538 \text{ per month.}}$$

General formula

$$B_k = P\,(F/P,\ i,\ k) - A\,(F/A,\ i,\ k), \qquad A_{\text{new}} = B_k\,(A/P,\ i_{\text{new}},\ N-k).$$

where B_k = balance still owed after k payments; $k = Y_{\text{term}} \times m$ (payments made during the first term); A = the original payment from part (a); $i = r_{\text{mort}}/m$ (old periodic rate); $i_{\text{new}} = r_{\text{new}}/m$ (new periodic rate after the term); $N - k$ = remaining payments over the rest of the amortization.

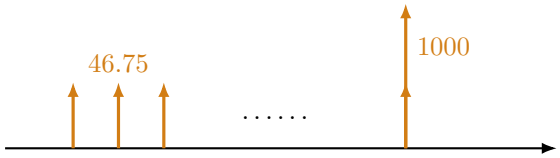
▶ Intuition: the payment rises from about \$2,338 to about \$2,538 because the interest rate went up from 5% to 6% while a large balance is still outstanding.

Bonds

- **Face value:** the amount of payment due upon maturity.
- **Coupon rate:** the rate used to calculate the coupon payments.
- **Frequency** of coupon payments.

The price is the present value of the coupons (an annuity) plus the face value:

$$P = A\,(P/A, i^*, N) + F\,(P/F, i^*, N).$$



A bond pays periodic coupons (A) plus the face value (F) at maturity.

Here y is the **yield**: it reflects the time value of money for the bond payments, and $i^* = y/m$.

General formula

$$P = A\,(P/A,\ i^*,\ N) + F\,(P/F,\ i^*,\ N), \qquad A = \frac{r_c}{m}\,F, \qquad i^* = \frac{y}{m}, \qquad N = Y \times m.$$

where P = bond price (today); F = face value (paid at maturity); A = coupon payment per period; r_c = coupon rate (annual); y = yield (annual); i^* = periodic yield; m = coupon payments per year; Y = years to maturity; N = total coupon periods.

▶ What is m? It is the number of coupon periods per year, i.e. how often the coupon is paid (and the yield compounded). So a bond that pays semiannual coupons has $m = 2$, and the per period rate used in the factors is $i^* = y/m$.

If the yield gets higher, the price P gets lower, because the price is just the present value of a fixed set of future cash flows (the coupons and the face value). A higher yield discounts those future payments more heavily, so each one is worth less today, and the total present value (the price) falls. This is the standard inverse relationship between bond price and yield.

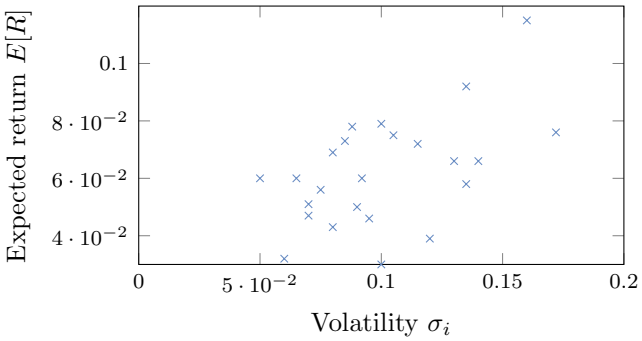
4 Week Four: Risk, Portfolios, CAPM, and Arbitrage

This part explains why we use CAPM (if you do not care, you can skip straight to the formula).

Financial risk and the need for a portfolio

Financial risk is the uncertainty of the future pay off. But when you plot the expected return against the volatility, there is no clear trend line.

Time Period (t)	Price (P)
0	20
1	21
2	22
3	20
4	21



A price series (left) and the expected return vs. volatility cloud (right): no clear trend.

For the price data, the per period (log) return and its vector and variance are:

$$P_{t_j} = P_{t_{j-1}}\,e^{\,r_{t_j}\Delta t}, \qquad r_{t_j} = \frac{1}{\Delta t}\ln\frac{P_{t_j}}{P_{t_{j-1}}}, \qquad \vec{R}_i = \begin{bmatrix} r_{t_1} \\ r_{t_2} \\ \vdots \\ r_{t_m} \end{bmatrix}, \qquad \sigma_i^2 = \mathrm{Var}(\vec{R}_i).$$

Modern Portfolio Theory (MPT)

So we use **MPT**: we want to maximize our return while minimizing the volatility. Assume we can invest our wealth by an amount x_i in the i-th asset. The expected return equals the weighted sum of the stocks we invested in, while the portfolio volatility is defined through the covariance matrix:

$$\vec{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{bmatrix},$$

where $\sigma_{ii} = \mathrm{Var}(\vec{R}_i)$ and $\sigma_{ij} = \sigma_{ji} = \mathrm{Cov}(\vec{R}_i, \vec{R}_j)$. Then

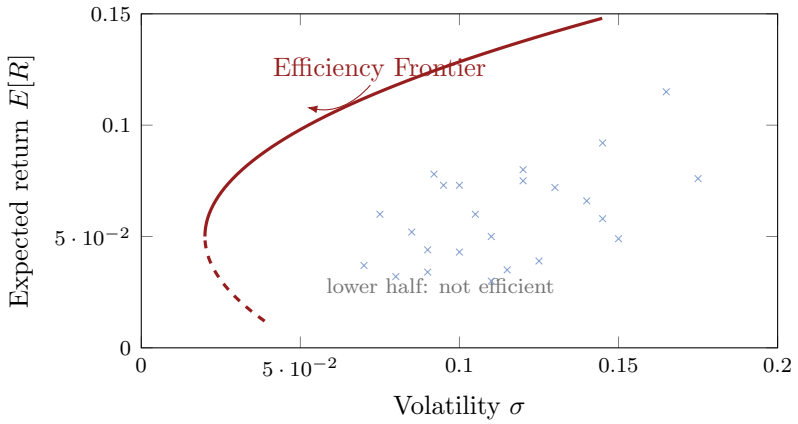
$$\sigma_p = \sqrt{\vec{X}^T \Sigma \vec{X}}, \qquad \text{optimize: maximize } \mathbb{E}\left[\vec{R}_p\right] \text{ for a given } \sigma_p.$$

We solve for X by maximizing the expected return of the portfolio $\mathbb{E}[R_p]$ for a given σ_p, which leads to CAPM. CAPM has a lot of limitations but gives us a weighted estimate of the price of risk.

Efficiency frontier

Using the assumptions and optimizing the expected return, we can create a portfolio of stocks for any given portfolio volatility, which leads to an **efficiency frontier**.

▶ What is the "high side"? For a given level of volatility there are two portfolios on the boundary, one with a high return and one with a low return. Only the upper (high return) half of the curve is interesting to us, because for the same risk you would always prefer the higher return. The lower half is dominated, so people mean the upper branch when they say "efficient frontier".



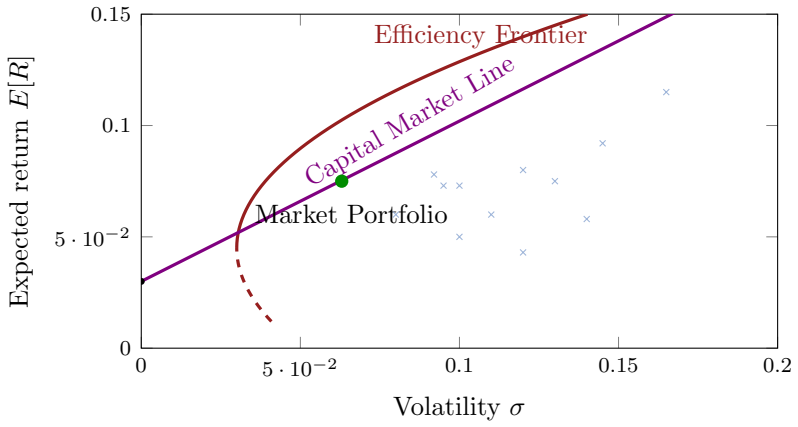
The frontier is the left boundary of what portfolios can reach. Only the solid upper branch (the "high side") is efficient; the dashed lower branch is dominated.

Capital Market Line

If it is possible to borrow or invest at the risk free rate, we can draw a tangent line that intersects the y axis at the risk free rate (the risk free rate has a volatility of 0).

- Where it touches the frontier: the **tangency portfolio** / **market portfolio**.
- The tangent itself: the **Capital Market Line (CML)**.

If you invest, you should expect your result to lie on this line.



The CML runs from r_f (on the y axis) and just touches the frontier at the market portfolio.

Worked example (leverage along the CML).

▶ This is the example that was hard to read on the slide, explained step by step.

Suppose the market portfolio has an expected return of 7% and the risk free rate is $r_f = 3\%$. You have \$100 of your own money, but instead of just buying the market portfolio you *leverage up*:

- Invest your \$100 in the market portfolio (MP).
- Borrow another \$50 at the risk free rate r_f and invest that in the MP as well.

Now you hold \$150 of the MP but only \$100 is yours, so you have leveraged up along the CML by a factor of $150/100 = 1.5$. Both the risk and the extra return scale by that same factor:

$$\sigma_p = 1.5 \times \sigma_{mp} = 1.5 \times 0.06 = 0.09, \qquad \mathbb{E}[R_p] = 1.5\,(7\% - 3\%) + 3\% = 9\%.$$

So by borrowing at r_f and overinvesting in the market, you move up and to the right along the CML: more volatility (0.09 instead of 0.06) in exchange for more expected return (9% instead of 7%).

General formula

$$\sigma_p = L\,\sigma_{mp}, \qquad \mathbb{E}[R_p] = L\big(\mathbb{E}[R_{mp}] - r_f\big) + r_f, \qquad L = \frac{\text{total invested}}{\text{own money}}.$$

where L = leverage factor ($L > 1$ means you borrowed, $L < 1$ means you also hold some risk free asset); σ_{mp} = volatility of the market portfolio; $\mathbb{E}[R_{mp}]$ = expected return of the market portfolio; r_f = risk free rate.

CAPM

- **MP** (market portfolio): represents the whole market.
- An asset has both **systematic** risk (associated with the market as a whole) and **idiosyncratic** risk (particular to the company).

- The market risk is designated by beta.

$$\mathbb{E}[R_i] = r_f + \beta_i\big(\mathbb{E}[R_{MP}] - r_f\big), \qquad \beta_i = \frac{\sigma_{i,MP}}{\sigma^2_{MP}} = \frac{\rho_{i,MP}\,\sigma_i}{\sigma_{MP}}.$$

General formula

$$\mathbb{E}[R_i] = r_f + \beta_i\big(\mathbb{E}[R_{MP}] - r_f\big), \qquad \beta_i = \frac{\sigma_{i,MP}}{\sigma^2_{MP}} = \frac{\rho_{i,MP}\,\sigma_i}{\sigma_{MP}}.$$

where $\mathbb{E}[R_i]$ = expected return the asset should earn; r_f = risk free rate (return for taking no risk); $\mathbb{E}[R_{MP}]$ = expected return of the whole market; $\big(\mathbb{E}[R_{MP}] - r_f\big)$ = market risk premium (extra return the market pays for its risk); β_i = how much of that market risk the asset carries ($\beta = 1$ moves with the market, $\beta > 1$ more volatile, $\beta < 1$ less); $\sigma_{i,MP}$ = covariance of the asset with the market; σ^2_{MP} = market variance; $\rho_{i,MP}$ = correlation of the asset with the market.

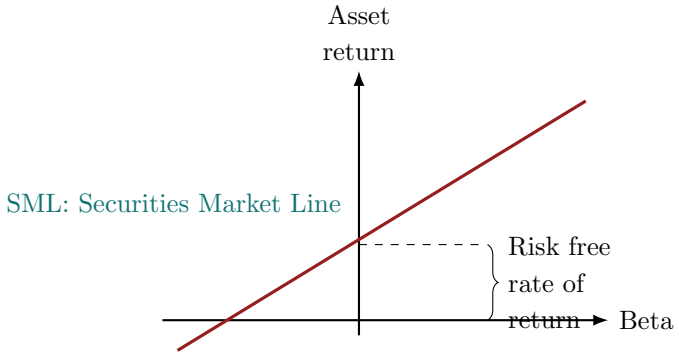
In words, the company return is

$$R_{C,t} = \alpha_C + \beta'_C R_{MP,t} + \varepsilon_{C,t}.$$

Volatility in $R_{MP,t}$ is systematic ("market") risk; volatility in $\varepsilon_{C,t}$ is idiosyncratic ("firm specific") risk. Investors can use portfolios to "diversify away" idiosyncratic risk, therefore there is no "value" to idiosyncratic risk. CAPM then says

$$\mathbb{E}[R_C] = r_f + \beta_C\big(\mathbb{E}[R_{MP}] - r_f\big),$$

where β_C is a measure of risk for the company.



The Securities Market Line: intercept at r_f, slope set by the market risk premium.

Example: Estimating future stock price

If asked for a future price, since we are already using CAPM (which carries a lot of error), you can use either a continuous or an effective rate; either way is fine.

A company with a beta of 0.9 has a stock price currently trading at \$170 a share. The current risk free rate is 2%, and the market rate is 7%. What is the expected price of a share 5 years from now, if the company pays no dividends?

$$\mathbb{E}[R_C] = r_f + \beta_C\big(\mathbb{E}[R_{MP}] - r_f\big) = 2\% + 0.9(7\% - 2\%) = 6.5\%.$$

$$P_t = P_0\, e^{R_C\, t} \quad \Longrightarrow \quad P_5 = \$170\, e^{0.065\times 5} = \$235.$$

General formula

$$\mathbb{E}[R_C] = r_f + \beta_C\big(\mathbb{E}[R_{MP}] - r_f\big), \qquad P_t = P_0\, e^{\mathbb{E}[R_C]\,t}.$$

where P_0 = current share price; P_t = expected share price at time t; t = number of years ahead; β_C = company beta; $\mathbb{E}[R_C]$ = the return demanded by CAPM, used as the growth rate of the price (continuous form shown; the effective form $P_t = P_0(1+\mathbb{E}[R_C])^t$ works too).

Example: Dividend paying stock

$P = \$10$, dividend $= \$0.10$ (quarterly), $\mathbb{E}[R_{MP}] = 10\%$, $r_f = 3\%$, $\beta = 1.2$, find P_3.

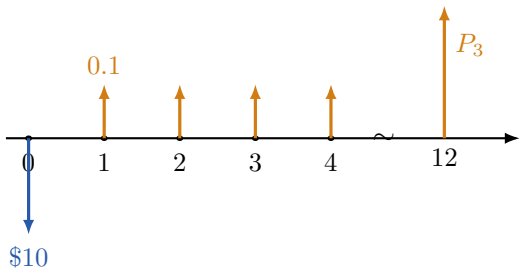
$$\mathbb{E}[R_C] = 3\% + 1.2(10\% - 3\%) = 11.4\%.$$

Convert to a quarterly company rate:

$$(1 + R_{C,q})^4 = (1 + 11.4\%) \quad \Longrightarrow \quad R_{C,q} = 2.736\%.$$

Cash flow: \$10 out today, \$0.10 each quarter for 12 quarters, and sell for P_3 at the end of year 3:

$$\$10 = \$0.10\,(P/A,\ 2.736\%,\ 12) + P_3\,(P/F,\ 11.4\%,\ 3) \quad \Longrightarrow \quad P_3 = \$12.43.$$



Dividends come in each quarter, the resale price P_3 at the end, and \$10 is paid today.

General formula

$$P_0 = D\,(P/A,\ i_q,\ n) + P_T\,(P/F,\ \mathbb{E}[R_C],\ T), \qquad (1+i_q)^m = 1 + \mathbb{E}[R_C].$$

where P_0 = price paid today; D = dividend per period; n = number of dividend periods; i_q = per period (e.g. quarterly) company rate; m = periods per year; $\mathbb{E}[R_C]$ = annual CAPM return; P_T = resale price after T years.

Arbitrage

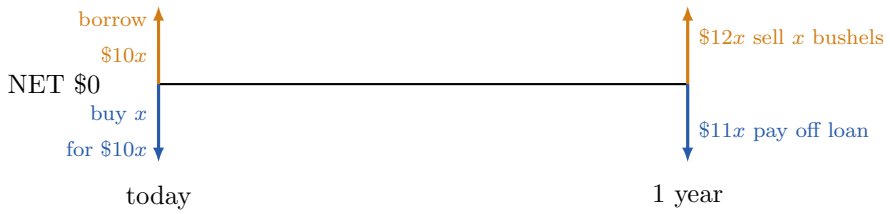
In financial markets: **arbitrage** is the practice of taking advantage of a price difference between two or more markets. Academic use: arbitrage occurs when there is an opportunity to achieve a risk free gain at a rate greater than the risk free rate. Fundamental to finance theory is the **no arbitrage assumption** ("no free lunch"). It works well in theory but not always in the real world.

Arbitrage Example

Say you could buy a bushel of corn for \$10 today and enter into a forward contract to deliver the corn for an agreed price of \$12 per bushel one year from now (assume it costs you nothing to store the corn). Say you could also borrow at 10%. What would you do?

Strategy (net \$0 today):

- Borrow $\$10x$.
- Buy x bushels for $\$10x$.



Borrow and buy today (net \$0), then in one year sell at $\$12x$ and repay $\$11x$, keeping $\$1x$.

In one year you sell for $\$12x$ and repay $\$11x$, so you keep $\$1x$ at no risk. As $x \to \infty$ this gives infinite profit. As demand for bushels today increases, the price will increase until there is no arbitrage. The no arbitrage price is

$$\frac{\$12}{1+10\%} = \$10.91.$$

General formula

$$S_0 = \frac{F}{1+r}, \qquad \text{equivalently} \qquad F = S_0\,(1+r).$$

where S_0 = fair spot (today) price with no arbitrage; F = agreed forward price; r = financing (borrowing) rate over the period. If the real spot price is below S_0, you can borrow, buy, and lock in a risk free profit.

Interest rate forwards

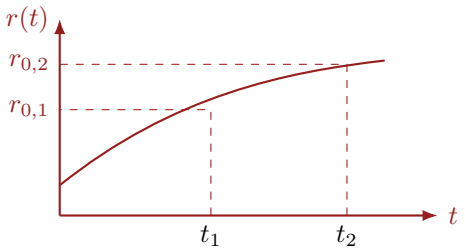
In this course we treat futures = forwards. At a given interest rate curve, the forward rate r_{t_1,t_2} is the appropriate rate between t_1 and t_2:

$$r_{1,2} = \frac{r_{0,2}\,t_2 - r_{0,1}\,t_1}{t_2 - t_1}.$$

General formula

$$r_{t_1,t_2} = \frac{r_{0,t_2}\,t_2 - r_{0,t_1}\,t_1}{t_2 - t_1}.$$

where $r_{0,t}$ = spot rate from today out to time t (read off the curve); r_{t_1,t_2} = forward rate that applies between t_1 and t_2; t_1, t_2 = the two times.



A rising interest rate curve, with the spot rates $r_{0,1}$ and $r_{0,2}$ marked.

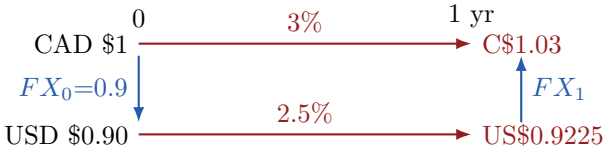
Foreign Exchange Example

Canadian risk free rate is 3%, US risk free rate is 2.5%, current FX = 0.9 US$/C$. What would be a fair FX rate one year from now that you would commit to today (the one year forward FX rate)?

Grow $1 CAD at 3% and $0.90 USD at 2.5%, then take the ratio:

$$\text{C\$1} \to \text{C\$1.03}, \qquad \text{US\$0.90}(1+0.025) = \text{US\$0.9225},$$

$$FX_1 = \frac{\text{US\$0.9225}}{\text{C\$1.03}} = 0.896 \text{ US\$/C\$}.$$



Grow each currency at its own risk free rate, then take the ratio to get the forward rate.

General formula

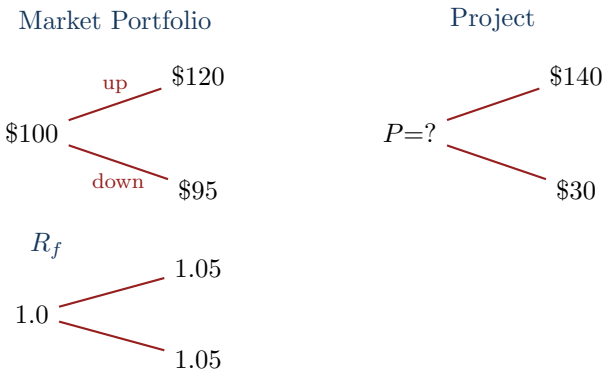
$$FX_1 = FX_0 \frac{1 + r_{\text{quote}}}{1 + r_{\text{base}}}.$$

where FX is quoted as (quote currency) per (base currency), here US$ per C$; FX_0 = current (spot) exchange rate; FX_1 = one year forward rate; r_{base} = risk free rate of the base currency (C$); r_{quote} = risk free rate of the quote currency (US$).

Replicating a project with the market

Project payoff is $140 if the market goes up and $30 if the market goes down. The risk free rate is 5% for this period. The current MP is priced at $100. If the market goes up it will be worth $120; if it goes down it will be worth $95.

We know the market portfolio (which can go up and down) and we know the project, but not P (the current price). The risk free asset is risk free, so either way it returns 1.05.



The market, the project, and the risk free asset as single step trees.

We replicate the project with a units of the MP and b units of the risk free asset ($a \sim MP$, $b \sim R_f$):

$$\begin{cases} \text{up:} & \$120\,a + \$1.05\,b = \$140, \\ \text{down:} & \$95\,a + \$1.05\,b = \$30. \end{cases} \quad \Longrightarrow \quad a = 4.4, \quad b = -369.53.$$

Therefore the no arbitrage price is

$$P = \$100 \times 4.4 - \$1 \times 369.53 = \$70.48.$$

General formula

$$\begin{cases} MP_u\,a + R_f\,b = X_u, \\ MP_d\,a + R_f\,b = X_d, \end{cases} \quad \Longrightarrow \quad P = MP_0\,a + b.$$

where a = units of the market portfolio in the replicating portfolio; b = dollars placed in the risk free asset ($b < 0$ means borrowing); MP_0 = current market price; MP_u, MP_d = market value in the up / down state; $R_f = 1 + r_f$; X_u, X_d = project payoff in the up / down state; P = the replicating cost, i.e. the no arbitrage price of the project.

Using probabilities. In the replication above we did not use the probability of the market going up or down. If the probability of the market going up is 60%, what is the expected return of the MP?

$$\mathbb{E}[R_{MP}] = \frac{\mathbb{E}[\text{payoff}_{MP}]}{\text{price}_{MP}} - 1 = \frac{\$120 \times 60\% + \$95 \times 40\%}{\$100} - 1 = 10\%.$$

And for the project:

$$\mathbb{E}[R_{proj}] = \frac{\$140 \times 60\% + \$30 \times 40\%}{\$70.48} - 1 = 36.2\%.$$

We can then use CAPM to find the beta of the project:

$$\mathbb{E}[R_{proj}] = r_f + \beta_{proj}\big(\mathbb{E}[R_{MP}] - r_f\big), \qquad 36.2\% = 5\% + \beta_{proj}(10\% - 5\%) \implies \beta_{proj} = 6.24.$$

What if the expected payoff of the project were \$140 regardless of the market?

$$\frac{\$140}{1.05} = \$133.$$

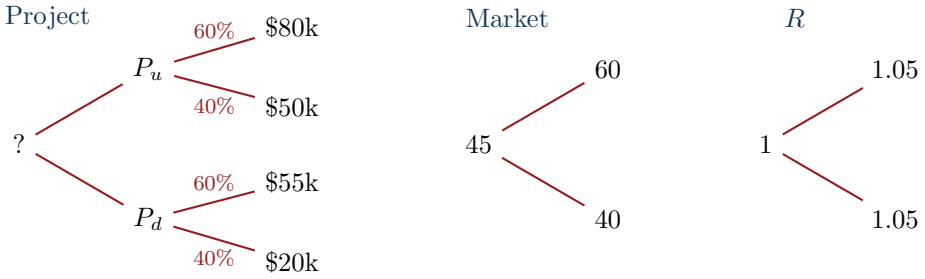
What if the project payoff had a 50% chance of being \$150 and a 50% chance of being \$130?

$$\mathbb{E}[\text{payoff}] = \$150 \times 50\% + \$130 \times 50\% = \$140.$$

For the 50/50 case, since there is no correlation with the market the beta is 0, so the price is the same as above, \$133.

Investments with systematic and idiosyncratic risk

- If the market goes up: 60% chance the payoff is \$80k, otherwise \$50k.
- If the market goes down: 60% chance the payoff is \$55k, otherwise \$20k.
- Current market portfolio is \$45 (up $\to$ \$60, down $\to$ \$40).
- Risk free rate is 5%.
- An initial investment of \$20k is required for R&D; determine the maximum price for the project.



The project has a two level tree: first the market (systematic), then the firm specific (idiosyncratic) branch.

First collapse the idiosyncratic branches into expected payoffs in each market state:

$$\mathbb{E}[P_u] = 0.6 \times 80\text{k} + 0.4 \times 50\text{k} = 68\text{k}, \qquad \mathbb{E}[P_d] = 0.6 \times 55\text{k} + 0.4 \times 20\text{k} = 41\text{k}.$$

Now replicate with a (market) and b (risk free):

$$\begin{cases} a(60) + b(1.05) = 68\text{k}, \\ a(40) + b(1.05) = 41\text{k}, \end{cases} \quad \Longrightarrow \quad a = 1.35\text{k}, \quad b = -12.38\text{k}.$$

Therefore

$$P_0 = 1.35\text{k} \times 45 - 12.38\text{k} = 48.37\text{k}.$$

However, since we need an initial investment of \$20k, we should buy the project for \$28.37k or less (that is, 48.37k − 20k).

General formula

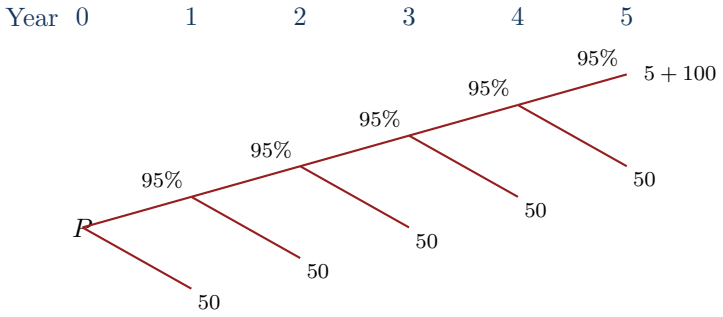
$$\mathbb{E}[P_u] = \textstyle\sum_k p_k\, X_{u,k}, \quad \mathbb{E}[P_d] = \sum_k p_k\, X_{d,k}, \quad P_0 = MP_0\, a + b, \quad P_{\max} = P_0 - I_0.$$

where $X_{u,k}, X_{d,k}$ = the idiosyncratic payoffs in the up / down market state; p_k = their probabilities; $\mathbb{E}[P_u], \mathbb{E}[P_d]$ = expected payoff once the firm specific risk is averaged out; a, b = replicating amounts (market and risk free), solved as before; I_0 = required initial investment (here the \$20k R&D); $P_{\max}$ = the most you should pay for the project.

Risky bond, accounting for default

A company is looking to issue a 5 year bond with a coupon rate of 5% paid annually. The risk neutral probability of the company defaulting on any given year is 5%, and in the case of a default only half of the principal will be paid out. What should be the price of the bond if the risk free rate is 3%? What is the yield rate?

Each year the bond either survives (95%) and pays its \$5 coupon, or defaults (5%) and pays \$50 (half of \$100 principal). At maturity a surviving bond pays $5 + 100$.



Each year: survive (95%) and pay the \$5 coupon, or default (5%) and recover \$50.

Work backwards. The value in the up (survive) state at $t = 4$ is:

$$P_{up}^{(4)} = 5 + \frac{105 \times 95\% + 50 \times 5\%}{1.03} = 104.27, \qquad P_{00}^{(4)} = 50 \text{ (default state)}.$$

Then roll the same formula back through each earlier stage:

$$P_{up}^{(3)} = 5 + \frac{95\% \times 104.27 + 50 \times 5\%}{1.03} = 103.60,$$

$$P_{up}^{(2)} = 5 + \frac{95\% \times 103.60 + 50 \times 5\%}{1.03} = 102.48,$$

$$P_{up}^{(1)} = 5 + \frac{95\% \times 102.48 + 50 \times 5\%}{1.03} = 102.41,$$

$$P_0 = \frac{102.41 \times 95\% + 50 \times 50\%}{1.03} = 96.88.$$

Finally, find the yield:

$$96.88 = 5\,(P/A, y, 5) + 100\,(P/F, y, 5) \quad \Longrightarrow \quad y = 5.735\%.$$

General formula

$$P^{(k)} = C + \frac{(1-p)\,P^{(k+1)} + p\,R}{1 + r_f}, \qquad P_0 = C\,(P/A, y, N) + F\,(P/F, y, N).$$

where $P^{(k)}$ = bond value in the surviving state at year k (worked backwards, with the maturity payment $C + F$ included at $k = N$); C = coupon payment; p = default probability per year; R = recovery if default (here half the principal); r_f = risk free rate; F = face value; y = yield, found by matching the price P_0 in the last equation.

5 Week Five: Comparing Projects (Present Worth and Annual Worth)

Multiple projects (investments)

1. **Independent:** make the decisions separately (project by project).
2. **Mutually exclusive:** choosing one will exclude all the other options, therefore we need to choose the best one.

Related but not mutually exclusive: form project combinations that are mutually exclusive, then find the best combination.

What does a project cash flow look like

- First we have the **first cost:** a one time cost at the beginning (car example: the money to buy the car).
- Then we have some revenues or savings (car example).
- **O&M** (operation and maintenance): car insurance or fuel.
- **Salvage value:** selling the car.

Because time changes the value of money, we want to evaluate benefits against costs at different times, i.e. evaluate if it is worth it, using a **discount rate**.

MARR

MARR is the minimum acceptable rate of return. It is usually estimated from the rate of return of the best alternative (if I do not invest in this one, what do I get if I put it somewhere else?). We want the project's return to be more than MARR, and we assume you can always earn MARR on some other project.

Comparison methods

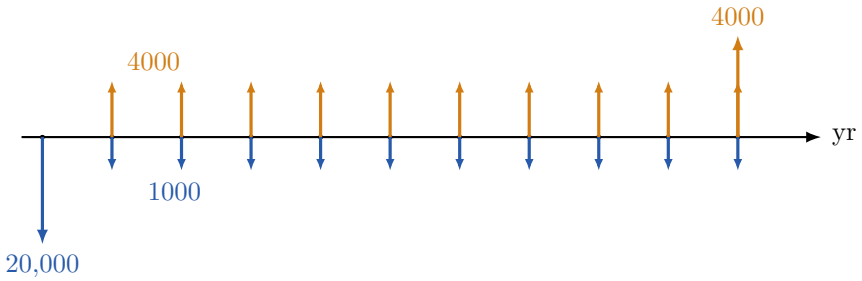
1. Present Worth (PW)

The **present worth** is the present value of the benefits minus costs, discounted at MARR. It is the amount by which the project is beating the best alternative, expressed in today's value.

- $PW > 0$: acceptable.
- $PW < 0$: not acceptable (this does not mean the project is unprofitable, but doing nothing is better).

Example 1: Calculating PW. You are planning to purchase a Honda Civic that has a life of 10 years. MARR is 5%. The annual benefits are worth \$4000, and the annual costs are worth \$1000. The first cost is \$20,000, and the salvage value after 10 years is \$4000. What is the PW of purchasing the Honda Civic?

$$PW = -20{,}000 + (4000 - 1000)\,(P/A, 5\%, 10) + 4000\,(P/F, 5\%, 10) = 5621.$$



Honda Civic: \$20,000 first cost at year 0, net \$3000 (benefit \$4000, cost \$1000) each year, and \$4000 salvage at year 10.

General formula

$$PW = -FC + (B - C)\,(P/A, i, N) + S\,(P/F, i, N), \qquad \text{accept if } PW > 0.$$

where FC = first cost (paid at year 0); B = annual benefit; C = annual cost; $(B - C)$ = net annual cash flow; S = salvage value at the end; i = MARR; N = project life.

2. Annual Worth (AW)

Same example, using **annual worth**: instead of collapsing everything to a present value, spread it into an equivalent level amount per year.

$$AW = -20{,}000\,(A/P, 5\%, 10) + 3000 + 4000\,(A/F, 5\%, 10) = 728.$$

(Check: this is the same decision as PW, since $AW = PW\,(A/P, 5\%, 10)$, and $5621\,(A/P, 5\%, 10) = 728$.)

General formula

$$AW = -FC\,(A/P, i, N) + (B - C) + S\,(A/F, i, N) = PW\,(A/P, i, N), \qquad \text{accept if } AW > 0.$$

where same symbols as PW above; the first cost is spread over the life with (A/P), and the salvage is spread with (A/F). Because PW and AW differ only by the factor $(A/P, i, N)$, they always give the same accept / reject decision.

Evaluating mutually exclusive projects

Here we can only choose one best option. Four steps to use PW to evaluate projects:

1. Define the time horizon.
2. Develop the cash flows for each alternative.
3. Calculate the PW using MARR.
4. Compare the PWs and pick the best.

General formula

choose the alternative $j^\star = \arg\max_j \ PW_j$ (all compared over the same horizon, and $PW_{j^\star} > 0$).

where PW_j = present worth of alternative j; every alternative must be evaluated over the *same* time horizon; if the best PW is still negative, do nothing. AW works the same way, comparing AW_j.

Example: same lives

Choosing between two mutually exclusive options:

	Honda Civic	Tesla Model S
FC	\$20,000	\$100,000
Annual benefit − cost	\$3,000	\$12,000
Life	10 years	10 years
Salvage Value	\$4,000	\$14,000
MARR	5%	5%

We calculated the Honda Civic already:

1. Both time horizons are 10 years.
2. Compute each PW over those 10 years.
3. $PW_{\text{Honda}} = 5621$.

And for the Tesla:

$$PW_{\text{Tesla}} = -100{,}000 + 12{,}000\,(P/A, 5\%, 10) + 14{,}000\,(P/F, 5\%, 10) = 1255.$$

Since the Honda Civic has a higher present worth ($5621 > 1255$), we choose that.

What if the time horizon is different?

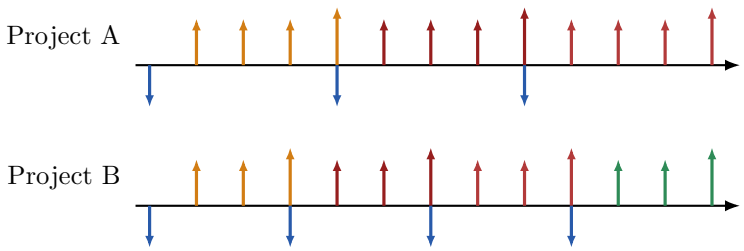
Example: one toothbrush costs \$5 and lasts two years, another costs \$4 and lasts one year. The two options cover different spans, so we cannot compare their present worths directly. There are two ways to fix this.

Method 1: Repeated lives

Assume the project repeats itself (e.g. you always need a car or a toothbrush). Use the **Least Common Multiple (LCM)** of the individual lives as the study horizon, then repeat each project's cash flow to fill that horizon.

Example 2: PW with repeated lives. Now compare the Honda Civic to a Lexus RX.

	Honda Civic	Lexus RX
FC	\$20,000	\$40,000
Annual benefit − cost	\$3,000	\$8,000
Life	10 years	**5 years**
Salvage Value	\$4,000	\$13,000
MARR	5%	5%



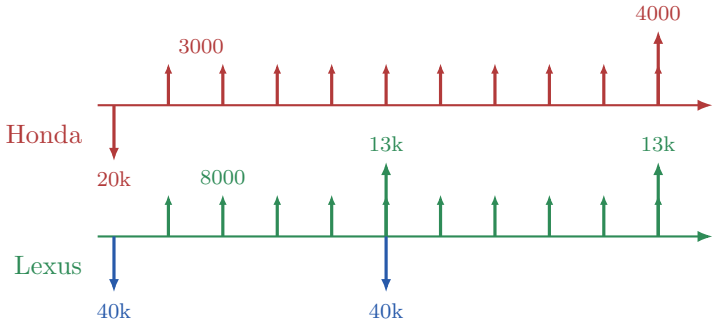
Repeated lives: each project is repeated (a down arrow restarts each cycle, colours mark the cycles) until both reach the LCM of their lives.

The LCM of 10 and 5 is 10 years. The Honda already spans 10 years ($PW = 5621$). For the Lexus, work out one 5 year cycle, then repeat it once more (shifted 5 years into the future):

$$PW_{\text{Lexus}}(5\text{ yr}) = -40{,}000 + 8000\,(P/A, i, 5) + 13{,}000\,(P/F, i, 5) = 4822,$$

$$PW_{\text{Lexus}}(10\text{ yr}) = 4822 + 4822\,(P/F, i, 5) = 8600.$$

Now we see the Lexus has a higher PW ($8600 > 5621$), so we choose the Lexus.



Honda over 10 years vs. the Lexus repeated twice (buy again at year 5) to fill the same 10 year horizon.

We can also use **AW** for repeated lives. The nice thing about AW is that it already annualizes, so you can compute each one over its *own* life and compare directly:

$$AW_{\text{Honda}} = 728, \qquad AW_{\text{Lexus}} = -40{,}000\,(A/P, 5\%, 5) + 8000 + 13{,}000\,(A/F, 5\%, 5) = 1114.$$

Again the Lexus wins ($1114 > 728$), the same decision as PW.

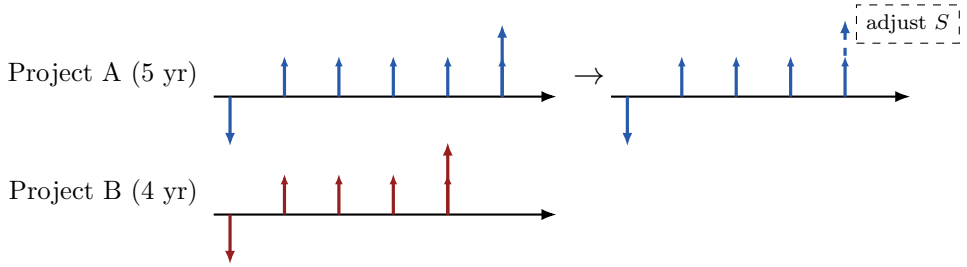
General formula

$$PW_{\text{LCM}} = PW_{\text{one life}} \sum_{k=0}^{n-1} (P/F, i, kN), \qquad n = \frac{\text{LCM}}{N}; \qquad \text{or just compare } AW \text{ over each own life.}$$

where N = one project's life; LCM = least common multiple of all the lives (the common horizon); n = number of times the project repeats; $PW_{\text{one life}}$ = present worth of a single cycle; the AW route avoids the repetition entirely, because AW is already a per year amount and does not change when the project repeats.

Method 2: Study period

Instead of stretching to the LCM, pick a common **study period** and cut every project off at the end of it. Here, if we terminate at the end of the shorter life, we compare both over that same period. When we cut a longer project short, we replace its salvage with an estimate of what it is worth at the end of the study period (adjust S).



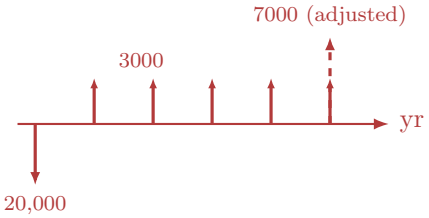
Study period: terminate the longer project (A) at the end of the shorter one (B), replacing its salvage with an adjusted value at that time.

We already know $PW_{\text{Lexus}}(5\text{ yr}) = 4822$. Now terminate the Honda at 5 years (instead of 10) and estimate its value at year 5 as \$7000 (the adjusted salvage), then compute its PW:

$$PW_{\text{Honda}}(5\text{ yr}) = -20{,}000 + 3000\,(P/A, 5\%, 5) + 7000\,(P/F, 5\%, 5) = -1527.$$

	Honda Civic	**Lexus RX**
FC	\$20,000	\$40,000
Annual benefit − cost	\$3,000	\$8,000
Life	**5** (was 10) years	**5 years**
Salvage Value	**\$7,000** (was \$4,000)	\$13,000
MARR	5%	5%

Since $PW_{\text{Lexus}}(5\text{ yr}) = 4822 > PW_{\text{Honda}}(5\text{ yr}) = -1527$, the Lexus wins under the study period method too.



Honda terminated at year 5 with an estimated \$7000 salvage, so it can be compared with the Lexus over the same 5 years.

General formula

$$PW = -FC + (B - C)\,(P/A, i, T) + S'\,(P/F, i, T),$$

where T = study period (a common horizon chosen for all alternatives, often the shorter life); S' = adjusted salvage, i.e. the estimated value of the asset at time T (not its value at the end of its full life); every other symbol is as before. Compare the PW (or AW) of each alternative over the same T and pick the best.